# 1. INTRODUCTION

This task is focused on improving the understanding of Supervised and Unsupervised Learning through Classification and Clustering algorithms. Interpreting the dataset with different types of classification and clustering algorithm can help us understand which model performed well in overall, these are calculated using different evaluation metrics. All these algorithms and evaluation metrics are explained based on the outcome of the dataset.

## 2. DATA EXPLORATION AND PREPROCESSSING

### 2.1 INITIAL EXPLORATION

The given Dataset has a total of 600 sample readings, 8 features and the Target variable: crop_health which is classified into 4 categories – Excellent, Good, Fair, and Poor. From the initial exploration all the 8 features are of numerical values and there are no missing values in the given dataset. The classes of target variables are evenly balanced with no major differences each class lies between 140 and 160.

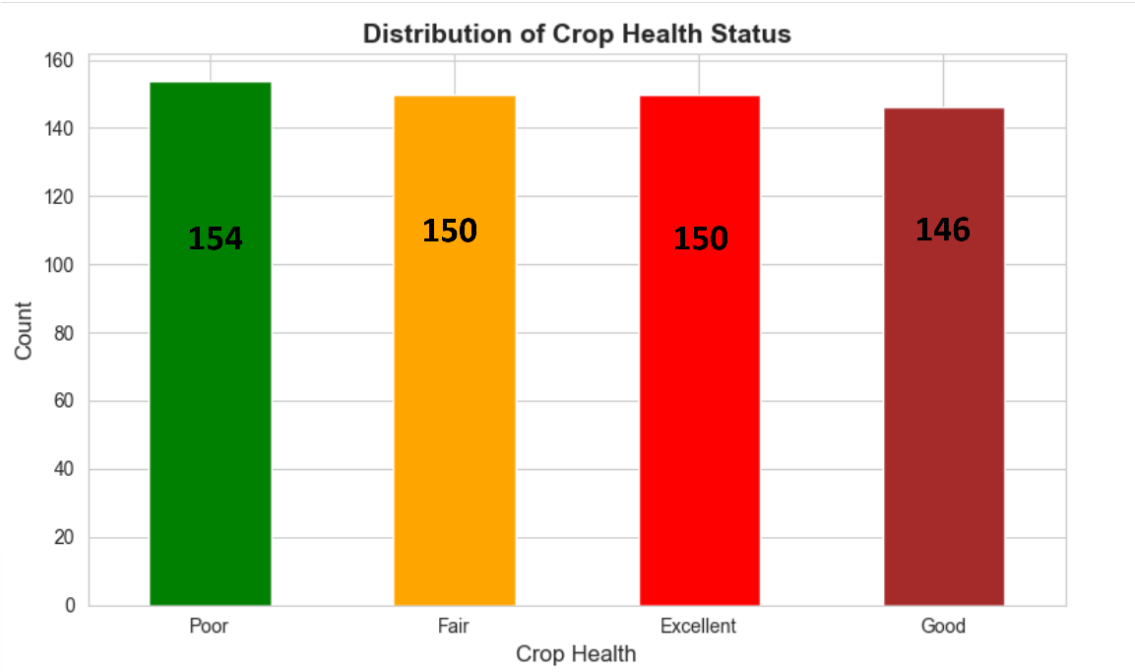


Fig 2.1.1

| | temperature | humidity | soil_moisture | pH_level | nitrogen | phosphorus | potassium | light_intensity |
|---|---|---|---|---|---|---|---|---|
| **count** | 600.000000 | 600.000000 | 600.000000 | 600.000000 | 600.000000 | 600.000000 | 600.000000 | 600.000000 |
| **mean** | 24.370233 | 68.003167 | 54.850500 | 6.085867 | 51.692167 | 23.617333 | 41.307667 | 5180.168333 |
| **std** | 3.823198 | 10.035435 | 11.028972 | 0.506953 | 16.261452 | 10.187689 | 12.338122 | 1802.584685 |
| **min** | 10.240000 | 30.000000 | 23.200000 | 5.000000 | 10.000000 | 5.000000 | 10.000000 | 1000.000000 |
| **25%** | 21.617500 | 62.500000 | 46.100000 | 5.750000 | 39.600000 | 16.250000 | 32.850000 | 3884.500000 |
| **50%** | 24.260000 | 68.750000 | 55.300000 | 6.070000 | 51.850000 | 22.400000 | 41.550000 | 5102.000000 |
| **75%** | 27.240000 | 75.125000 | 63.500000 | 6.380000 | 63.750000 | 30.200000 | 50.125000 | 6454.000000 |
| **max** | 35.000000 | 90.000000 | 80.000000 | 8.000000 | 100.000000 | 50.000000 | 80.000000 | 10000.000000 |

From the Summary table, count, mean, standard deviation, minimum value, maximum value, and 25, 50 & 70 % of each feature. From this the limits of the features can be understood.

Fig 2.1.2

Histogram represented a varied distribution in most of the features, like temperature, nitrogen, and potassium have normal distribution.

Humidity and Soil_moisture are left skewed.

pH_level and Phosphorus are right skewed.

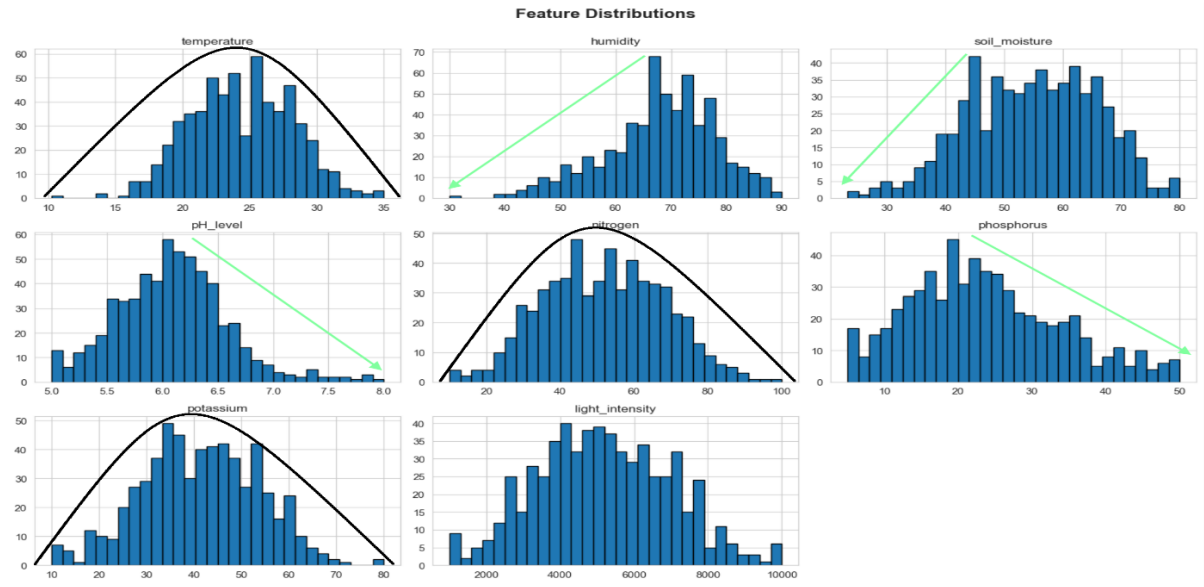


Fig 2.1.3

From the Boxplot it is visible that there are some outliers in some of the features like pH_level, nitrogen, and potassium but not major ones.

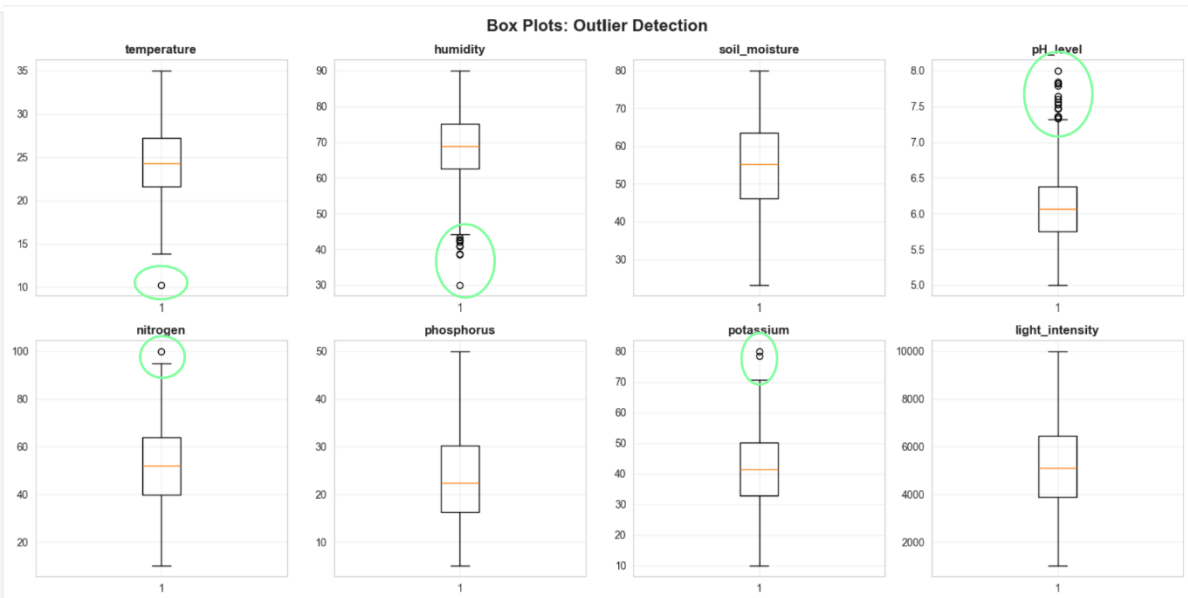


Fig 2.1.4

There is no highly correlated pair (|r| > 0.7).

The highest correlation in the correlation matrix is between:-

- Potassium and Soil _moisture = 0.51
- Temperature and Humidity = 0.47.

The lowest correlation is between:-

- Humidity and Potassium = -0.65.

Fig 2.1.5

The notable pattern from this dataset are:-

1. As the soil_moisture increases the content of Potassium will also increases.
2. Relation between Temperature and Humidity.
3. Relation between Nitrogen and Humidity.

## 2.2 DATA PREPROCESSING

Data Quality issues can be handled by fixing the issues with missing values and handling outliers. Since there is no missing data, it is sorted. For handling the Outliers present, Capping method is used because it will retain all data points and reduce the influence of extreme values. (Lomash Bhuva, 2025)

```
#Option 1: Capping (clip extreme values)
for col in df.columns[:-1]:
    q99 = df[col].quantile(0.99)
    q01 = df[col].quantile(0.01)
    df[col] = df[col].clip(lower=q01, upper=q99)
```

Fig 2.2.1

After performing Capping method, the final dataset shape is (600,9)

Dataset is divided into training set 75% and testing set 25% and stratified for maintaining balanced classes and random_state is fixed to 42 to ensure reproducibility, debugging and comparison of results. (saturncloud.io, 2023). After performing train_test_split, the Training set have 450 samples and Testing set have 150 samples.

```
Train-Test Split:
======================================
Training set: 450 samples
Test set: 150 samples

Training set class distribution:
crop_health
Poor         116
Fair         113
Excellent    112
Good         109
Name: count, dtype: int64

Test set class distribution:
crop_health
Poor         38
Excellent    38
Good         37
Fair         37
Name: count, dtype: int64
```

Fig 2.2.2

Feature Scaling have a significant impact on the whole dataset; all the features now have a mean approximatively around ‘0’ and Standard Deviation approximatively around ‘1’. This will help in the processing of data more accurately. It is visible from the boxplot after scaling.

```
Feature scaling completed!

Scaled feature statistics (training set):
      temperature  humidity  soil_moisture  pH_level  nitrogen  phosphorus  \
mean       0.0000    0.0000         0.0000    0.0000   -0.0000     -0.0000
std        1.0011    1.0011         1.0011    1.0011    1.0011      1.0011

      potassium  light_intensity
mean     0.0000           0.0000
std      1.0011           1.0011
```

Fig 2.2.3

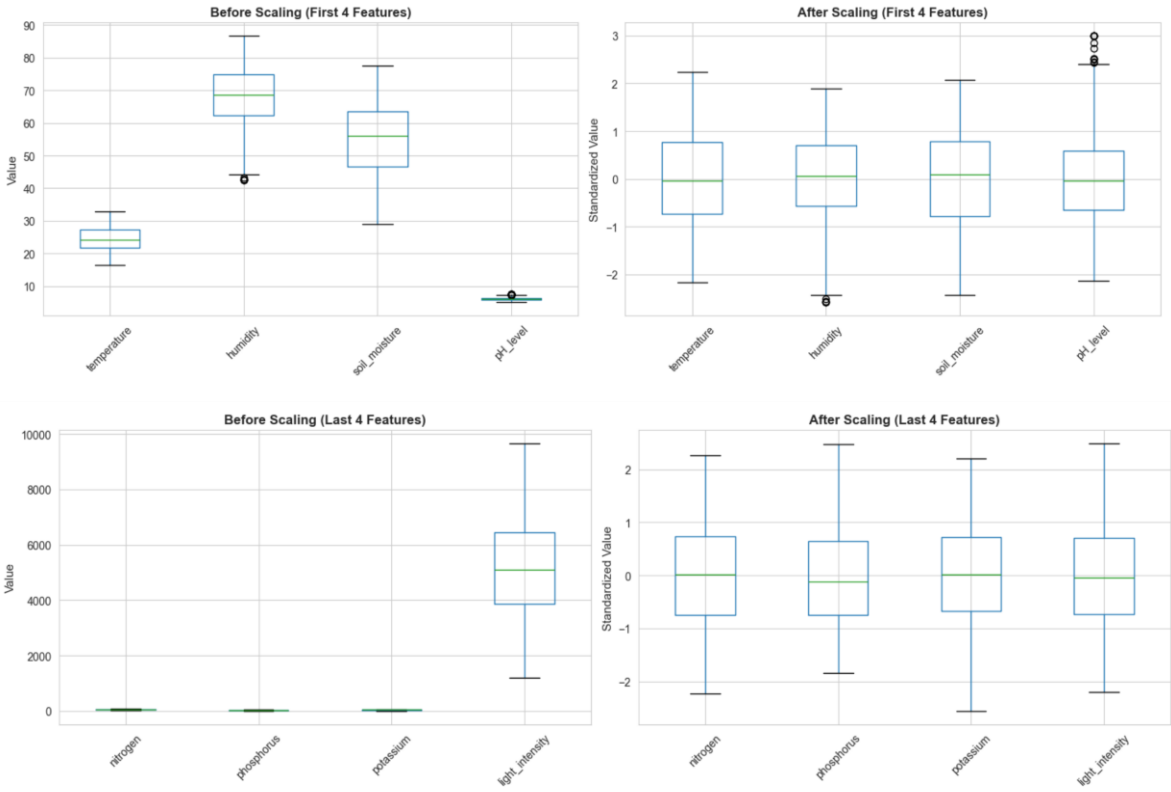


Fig 2.2.4

# 3. CLASSIFICATION ANALYSIS

## 3.1 BUILDING MULTIPLE CLASSIFIERS

Best model is selected based on the highest testing accuracy.

| Decision Tree Classifier | Naïve Bayes Classifier – Gaussian NB | K – Nearest Neighbours |
|---|---|---|
| Tired with multiple depths= [3, 4, 5, 7, 8, 9, 10, None] And the best test accuracy was obtained in depth of '4' with train_accurcay = 88.00 % and test_accuracy = 82.00 % | Performed Naïve bayes classifier and got a Training accuracy of 86.44 % and Test accuracy of 84.67 % | Tested the model with multiple k values [3, 4, 5, 7, 9,11] and the best performance was observed in k = 9.<br><br>Selected the optimal k = 9, because it was performing well in testing set. |
| Train_accuracy = 88 %<br><br>Test_accuracy = 82 % | Train_accuracy = 86.44 %<br><br>Test_accuracy = 84.67 % | Train_accuracy = 88.67 %<br><br>Test_accuracy = 88.00 % |

Table 3.1.1

## 3.2 EVALUATE AND COMPARE CLASSIFIERS

In this analysis, we have used three machine learning algorithms with its own benefits and capabilities.

| Decision Tree Classifier | Navie Bayes Gaussian NB | K-Nearest Neighbour |
|---|---|---|
| Accuracy= 82% | Accuracy= 84.67% | Accuracy= 88% |
| Interpretability= Can understand from the tree structure. | Interpretability= It is based on the class proportions | Interpretability= can understand more by plotting it against k values |

Table 3.2.1

```
Naive Bayes Classifier:
Training Accuracy: 0.8644
Test Accuracy: 0.8467
```

Fig 3.2.1

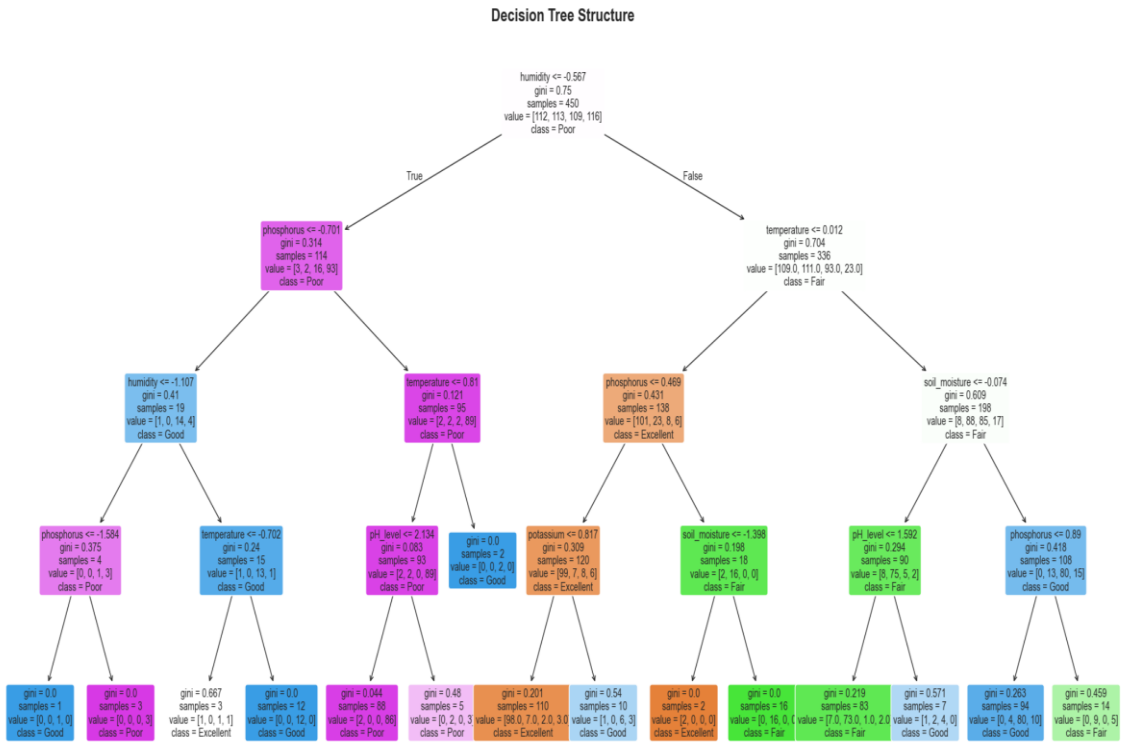


Fig 3.2.2

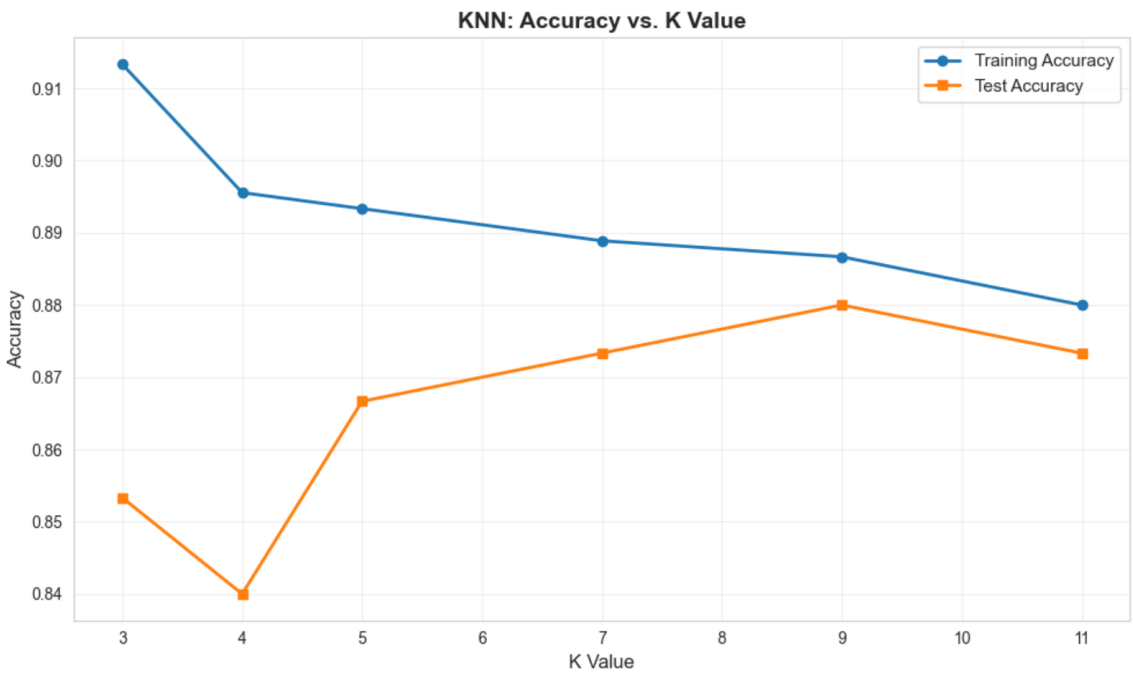


Fig 3.2.3

The best performed model is evaluated by the Confusion Matrix, Performance Metrics and Comparative Analysis.

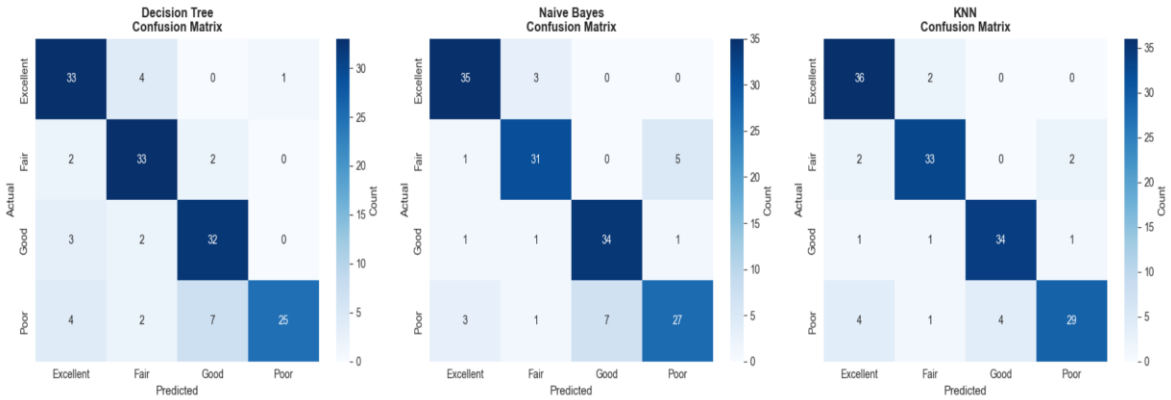


Fig 3.2.4

In the confusion Matrix fig(3.2.4):-

- Decision tree classifier = 123 correct predictions and 27 misclassifications.
- Naïve Bayes classifier = 127 correct predictions and 23 misclassifications.
- K-Nearest Neighbour = 132 correct predictions and 18 misclassifications.
- K- Nearest Neighbour performed effectively than other two models and K-Nearest Neighbour is followed by Naïve Bayes.

Additionally, there are some common misclassification patterns in all the three models:-

In Decision Tree, Naïve Bayes, and K-Nearest Neighbour classifier, Poor was misclassified as Good for 7, 7 and 4 times respectively.
Poor is misclassified as excellent in all three models for 4, 3, 4 times respectively.
In Naïve Bayes, fair is misclassified as poor for 5 times.

Conclusion:- Common confusion between “Poor vs Good” and “Poor vs Excellent”

```
Classification Model Comparison:
================================================================================
              Model  Accuracy  Precision (macro)  Recall (macro)  F1-Score (macro)
      Decision Tree    0.8200             0.8332          0.8208            0.8182
        Naive Bayes    0.8467             0.8459          0.8471            0.8448
K-Nearest Neighbors    0.8800             0.8825          0.8803            0.8790
================================================================================
```

Fig 3.2.5

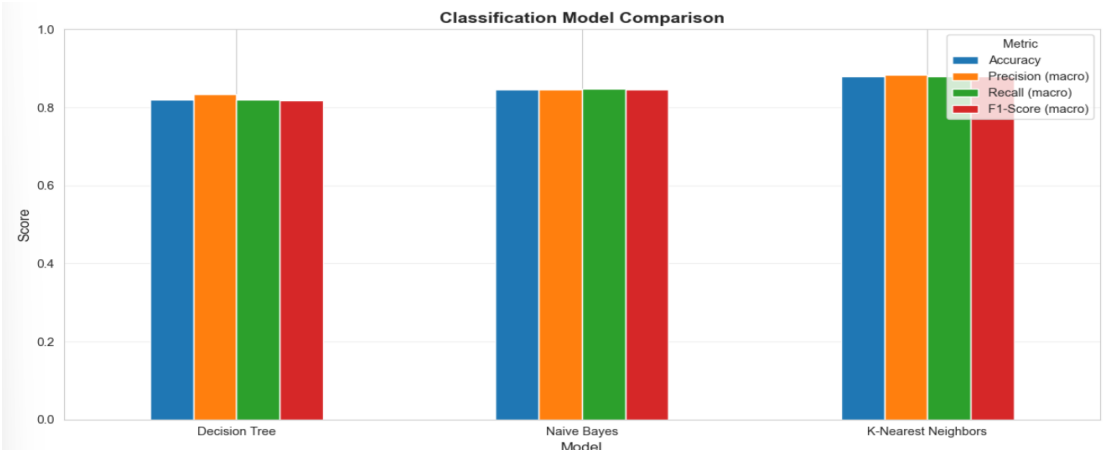


Fig 3.2.6

From the Performance Metrics and comparative analysis, we can conclude that KNN performed well as compared to all other models because, proved from performance metrics:-

- K-Nearest Neighbour, highest value in all metrics –

  Accuracy = 88.00 %

  Precision = 88.25 %

  Recall = 88.03 %

  F1-score = 87.90 %.

Conclusion:- KNN has performed best as compared to others.

The model which performed best for each class can be identified by considering the F1-score, which is the balanced measure (Harmonic mean of precision.)

Decision Tree:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Excellent | 0.79 | 0.87 | 0.82 | 38 |
| Fair | 0.80 | 0.89 | 0.85 | 37 |
| Good | 0.78 | 0.86 | 0.82 | 37 |
| Poor | 0.96 | 0.66 | 0.78 | 38 |
| accuracy | | | 0.82 | 150 |
| macro avg | 0.83 | 0.82 | 0.82 | 150 |
| weighted avg | 0.83 | 0.82 | 0.82 | 150 |

Naive Bayes:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Excellent | 0.88 | 0.92 | 0.90 | 38 |
| Fair | 0.86 | 0.84 | 0.85 | 37 |
| Good | 0.83 | 0.92 | 0.87 | 37 |
| Poor | 0.82 | 0.71 | 0.76 | 38 |
| accuracy | | | 0.85 | 150 |
| macro avg | 0.85 | 0.85 | 0.84 | 150 |
| weighted avg | 0.85 | 0.85 | 0.84 | 150 |

KNN:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Excellent | 0.84 | 0.95 | 0.89 | 38 |
| Fair | 0.89 | 0.89 | 0.89 | 37 |
| Good | 0.89 | 0.92 | 0.91 | 37 |
| Poor | 0.91 | 0.76 | 0.83 | 38 |
| accuracy | | | 0.88 | 150 |
| macro avg | 0.88 | 0.88 | 0.88 | 150 |
| weighted avg | 0.88 | 0.88 | 0.88 | 150 |

Fig 3.2.7

For the Fair, Good and Poor classes, KNN model is performed best, where F1-score = 89 %, 91 %, and 83% respectively.

For the Excellent class, Naïve Bayes Classifier performed best, where F1-score = 90 %.

Overall KNN has performed best in most of the classes other than Excellent class.

### 3.3 Model Interpretation

Humidity, Temperature, Soil moisture and Phosphorus are the highest performed features according to the Decision Tree Classifier algorithm. Temperature (24.9119%) and Humidity (27.2624%) have a significant role in the growth and development of plants, it determines the metabolism, photosynthesis, and other processes of plants. (logicsun, 2023). These features are followed by Soil_moisture at a percentage of 21.1845 %, soil_moisture is important for various chemical and physical processes. Potassium, Nitrogen, Light_intensity, and pH _level have its own importance but less as compared to Humidity, Temperature, Soil moisture and Phosphorus. But in this model Nitrogen and Light intensity is not showing any importance it is because Decision tree calculates feature importance according to the reduction in Gini index in this case as we can see in the decision tree Fig(3.2.2) and also this tree is a shallow tree. (GeeksforGeeks, 2024)

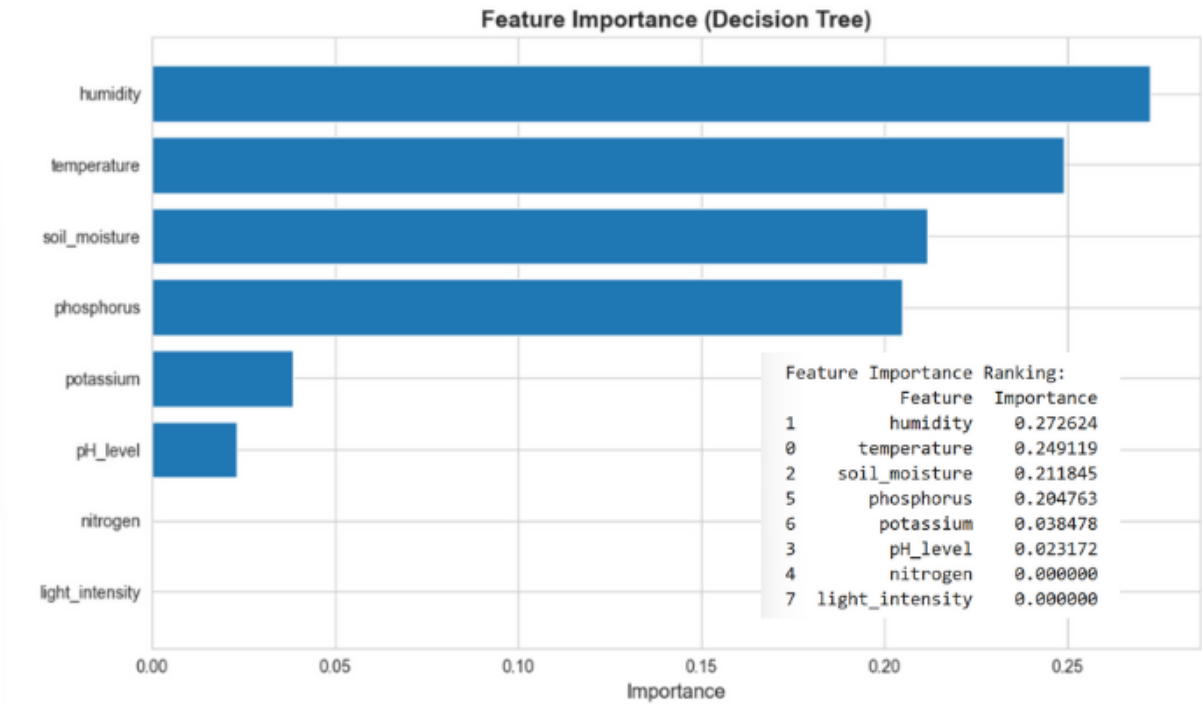


Fig 3.3.1

In this Dataset most of the misclassifications have occurred in “Poor vs Good”, and “Poor vs Excellent”.

These misclassifications have occurred due to variations in values of different parameters. For example, the sample in rows 213 and 358 its actual class is Excellent, but it is misclassified as Fair class even though it has high temperature and humidity, it might have occurred due to confusion between other values. It is because KNN model will always compare with the nearest neighbours and conclude.

```
Examples of misclassified instances:
     temperature  humidity  soil_moisture  pH_level  nitrogen  phosphorus  \
213        26.18      75.1           53.5      6.05      47.6        23.3
592        21.02      65.3           66.1      5.46      60.8        22.5
327        22.61      52.7           74.0      6.44      69.0         5.0
358        26.43      74.2           39.9      6.45      50.5        11.8
187        21.49      66.5           51.3      5.99      52.9        27.2
321        23.36      62.5           51.0      6.65      42.2        24.7
568        19.35      73.2           55.7      6.41      71.8        21.7
77         27.79      69.6           44.5      6.67      28.7        35.5
522        21.88      67.5           62.0      6.10      60.2        36.4
111        21.38      70.2           54.9      5.32      61.2        18.9

     potassium  light_intensity     Actual  Predicted
213       39.2           4544.0  Excellent       Fair
592       61.1           7590.0       Good       Poor
327       50.0           3134.0       Poor       Good
358       34.4           4442.0  Excellent       Fair
187       44.6           5345.0       Fair  Excellent
321       46.1           6447.0       Fair       Poor
568       40.1           4178.0       Poor  Excellent
77        36.8           7747.0       Poor       Fair
522       51.8           7851.0       Fair       Poor
111       32.0           3961.0       Good  Excellent
```

```
Total misclassified instances: 18
Percentage: 12.00%

Misclassification patterns:
Actual     Predicted
Poor       Good        4
           Excellent   4
Excellent  Fair        2
Fair       Excellent   2
           Poor        2
Good       Fair        1
           Excellent   1
           Poor        1
Poor       Fair        1
Name: count, dtype: int64
```

Fig 3.3.2

K-Nearest Neighbour compare the values with each other, and this can be improved by using different distance metrics like Euclidean distance, Manhattan distance, Minkowski distance and Hamming distance. (IBM, 2021)

In real world, farmers can get some insights from this dataset like the best crop category, soil, temperature, growing environment. But this model can also give some wrong information like poor misclassified as good and excellent and excellent misclassified as fair and vice versa. This can lead to wastage of crops, time and money. For example, the farmer might not consider or give more care to the crops with predicted fair class, but they are of Excellent class, this leads to decrease in harvesting.

# 4. CLUSTERING ANALYSIS

## 4.1 Determining Optimal Number of Clusters

For finding the optimal number of clusters, four evaluation metrics are used with K value range from 2 to 10.

In Elbow method, we are computing the WCSS for all K and plotting it against K and finding the elbow point.
Elbow point is the point where the improvement slows down. From K=4 is slowing down, So the number of clusters become K=4. (GeeksforGeeks, 2019)

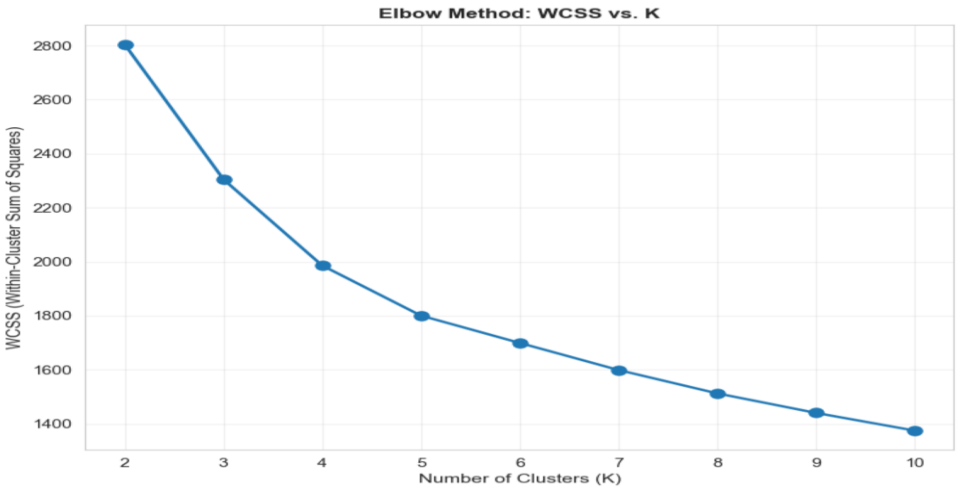


Fig 4.1.1

In Silhouette Score, the higher Silhouette Score is the optimal number of clusters can be identified. Here the maximum score is in K= 4.

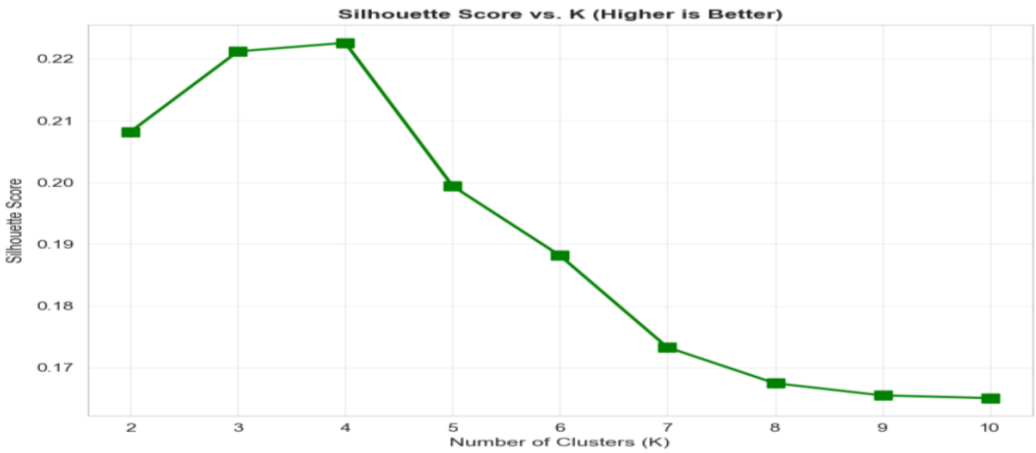


Fig 4.1.2

In Calinski-Harabasz index, the highest index is best for the optimal number of clusters K=2.

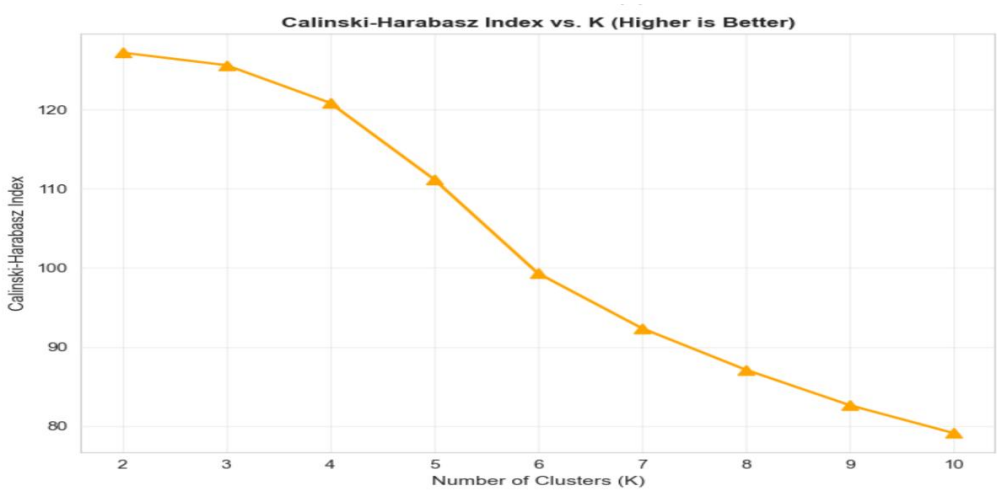


Fig 4.1.3

In Davies-Bouldin Index, the lowest index is best for the optimal number of clusters here it is K=4.

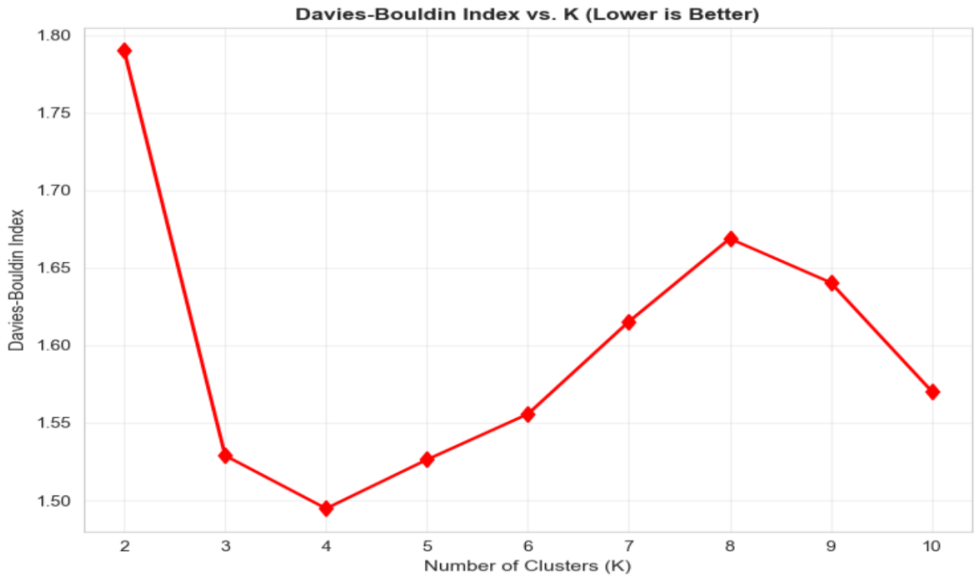


Fig 4.1.4

The disagreement occurs in the evaluation metrics of Calinski-Harabasz index, it's optimal number of clusters as 2, but other 3 metrics have the optimal number of clusters as 4 as seen in plotting. 3 / 4 of the metrics strongly suggested the number of clusters.

```
Optimal K Analysis Summary:
    K       WCSS  Silhouette  Calinski-Harabasz  Davies-Bouldin
0   2  2803.9351      0.2082           127.1916          1.7900
1   3  2304.7268      0.2212           125.6086          1.5289
2   4  1985.8388      0.2226           120.8416          1.4948
3   5  1800.4913      0.1994           111.1893          1.5263
4   6  1699.4164      0.1881            99.3116          1.5557
5   7  1599.2401      0.1733            92.3706          1.6153
6   8  1512.7303      0.1675            87.1247          1.6688
7   9  1440.4598      0.1655            82.6435          1.6405
8  10  1374.9485      0.1651            79.1159          1.5704
```

Fig 4.1.5

So, from the plotting and visualization, we can conclude that, the optimal number of clusters is 4 and it is supported by 3 evaluation metrics-Elbow Method – WCSS, Silhouette Score, Davies-Bouldin Index.

### 4.2 Appling Clustering Algorithms and Evaluation

Here three clustering algorithms are implemented and computed their evaluation metrics to understand the best performing unsupervised algorithm.

```
Clustering Method Comparison:
================================================================================
      Method  Silhouette  Calinski-Harabasz  Davies-Bouldin  Clusters  Noise Points
     K-Means      0.2226           120.8416          1.4948         4             0
Hierarchical      0.1898           103.8549          1.6043         4             0
      DBSCAN     -0.0769             7.1009          1.1087         7            55
```

Fig 4.2.1

Three evaluation metrics are used. Silhouette index and Calinski-Harabasz where higher values are best considered for better clustering, and Davies-Bouldin Index where the less value is preferred.

In the comparison table K-means is having a higher value in Silhouette and Calinski-Harabasz index and Davies-Bouldin having the 2nd lowest index.

But in DBSCAN which is a density-based algorithm, Silhouette and Calinski-Harabasz is having the smallest index but ideal in Davies-Bouldin index as compared to other methods.

Hierarchical method is having a balanced metrics values in all methods and it has not outperformed K-means in any metrics. From this comparison we can conclude that K-means clustering method performed overall well in all the metrics.

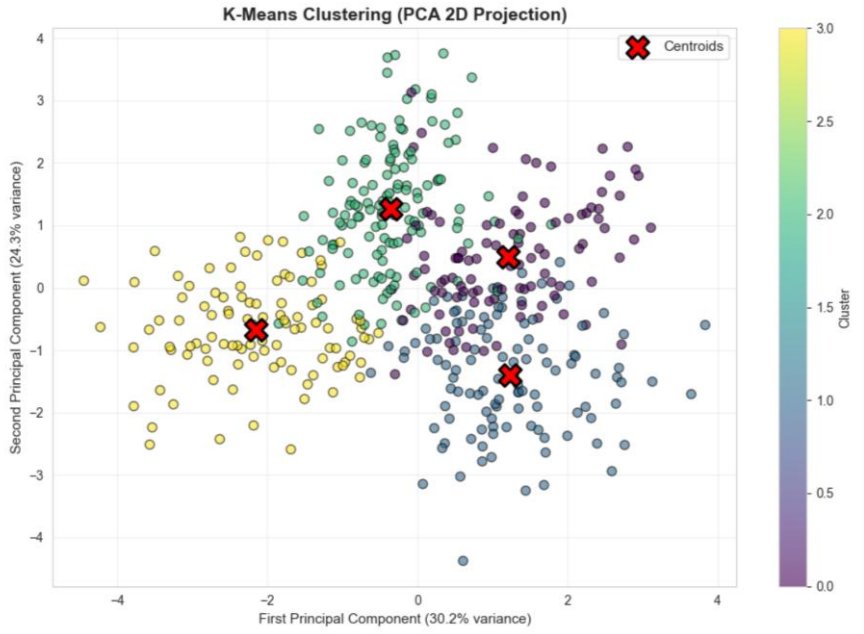


Fig 4.2.2

K-means clustering groups similar datapoints into each cluster and these clusters can help find hidden patterns and categorize data based on similarity later which would be useful in decision making like a filed in agriculture in our dataset. (GeeksforGeeks, 2017).

Each Clusters characteristics can be explained from the cluster profiles, Cluster vs True crop health Cross tabulation and from heat map.

```
K-Means Cluster Profiles:
================================================================================
         temperature  humidity  soil_moisture  pH_level  nitrogen  phosphorus  \
Cluster
0              0.795     0.944         -0.762     0.021    -0.295       0.590
1             -0.770     0.513         -0.728    -0.229     1.107      -0.636
2              0.644    -0.145          0.776    -0.086    -0.275      -0.480
3             -0.841    -1.355          0.553     0.348    -0.551       0.739

         potassium  light_intensity
Cluster
0           -0.791           -0.110
1           -0.719            0.381
2            0.612           -0.900
3            0.795            0.906
================================================================================

Cluster Sizes:
Cluster
0    104
1    111
2    135
3    100
```

Fig 4.2.3

```
Cluster vs. True Crop Health Cross-Tabulation:
True_Health  Excellent  Fair  Good  Poor
Cluster
0                   10    88     4     2
1                   95    11     2     3
2                    4     8    98    25
3                    3     6     5    86
```

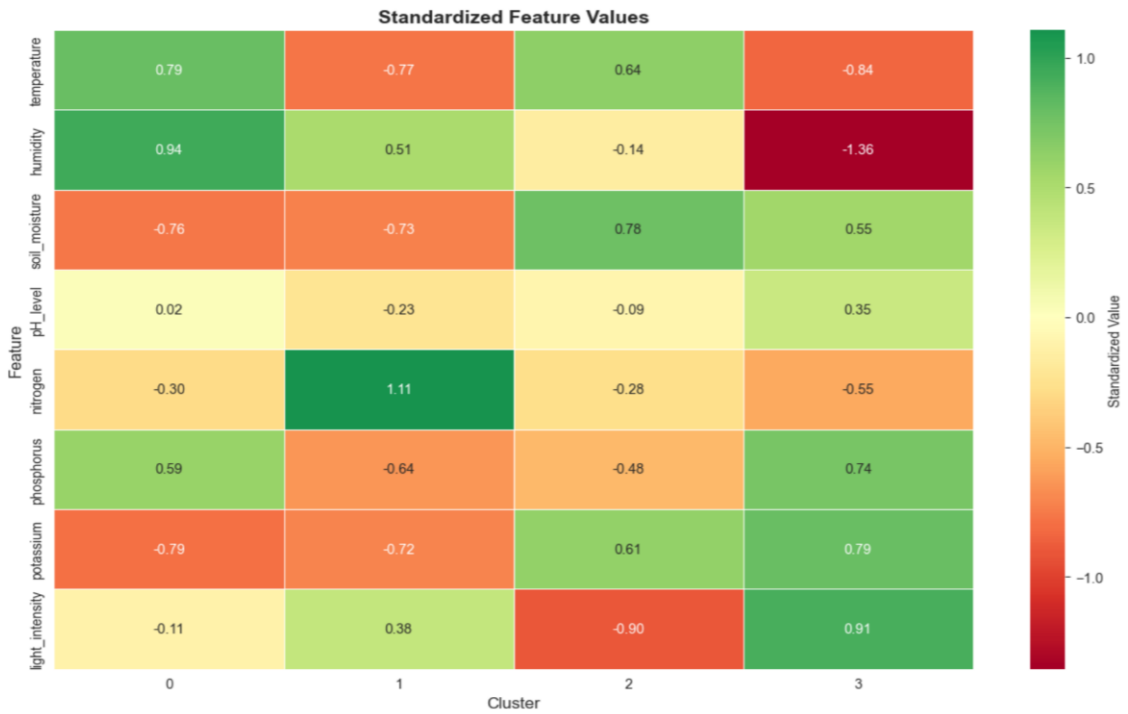


Fig 4.2.4

| Cluster 0 | Cluster 1 | Cluster 2 | Cluster 3 |
|---|---|---|---|
| High Temperature, Humidity | High nitrogen content, | High soil moisture, temperature, and potassium level. | All features except temperature, humidity and nitrogen are moderate to high. |
| Moderate-phosphorus<br><br>Neutral-pH level | Moderate humidity and light intensity, | | Moderately less nitrogen content. |
| Not too less nitrogen and light intensity | Low temperature, soil moisture, potassium and phosphorus | With humidity, pH level, nitrogen and phosphorus having a low-level content but not too deficient. | Low temperature, humidity |
| In cross tabulation, cluster 0 belongs to fair class | In cross tabulation, cluster 1 belongs to excellent class. Particularly because of the high nitrogen content. | In cross tabulation, cluster 2 belongs to good class., | In cross tabulation cluster 3 belongs to Poor class this is because despite having a good range of values in all other features other than temperature, humidity. Nitrogen makes it falls into Poor class. |
| 88 out of 104 categorized as fair in cluster 0. | 95 out of 111 categorized as Excellent in cluster 1 | 98 out of 135 categorized as Good in cluster 2 | 86 out of 100 categorized as Poor in cluster 3 |
| 84.62% of cluster 0 is matched as Fair. | 85.58% cluster 1 is matched as Excellent. | 72.59% cluster 2 is matched as Good. | 86% of the overall cluster 3 is matched as poor. |

Table 4.2.1

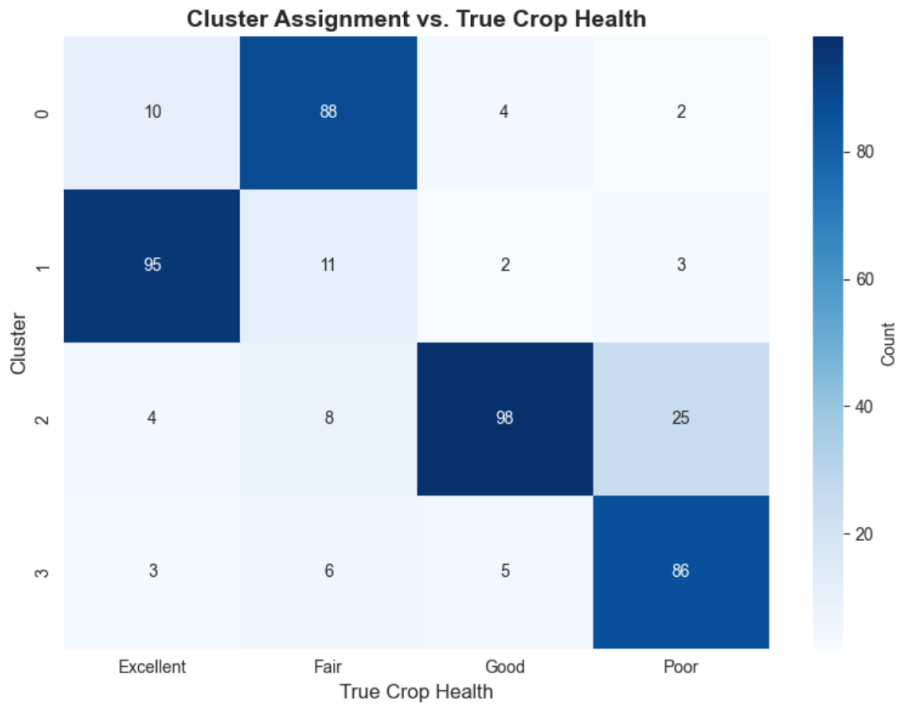


Fig 4.2.5

In overall, Clustering provided an 81.56% accuracy which means 81.56% are correctly matched to their corresponding classes.

Cluster 0, which belongs to class Fair, but it was misclassified as Excellent for 10 times and Good and Poor for 4 and 2 times respectively.

Cluster 1, which belongs to class Excellent, but it was misclassified as Fair for 11 times and Good and Poor for 2 and 3 times respectively.

Cluster 2, which belongs to class Good, but it was misclassified as Poor for 25 times and Excellent and Fair for 4 and 8 times respectively.

Cluster 3, which belongs to class Poor, but it was misclassified as Excellent, Fair and Good for 3,6, and 5 times respectively.

## 5.CONCLUSION

Supervised Learning and Unsupervised Learning was conducted in this dataset using different algorithms and was able to conclude that supervised learning particularly KNN achieved highest accuracy 88% as compared to unsupervised learning K-means Clustering algorithm, where it achieved 81.56% accuracy.

Supervised Learning uses labelled data, so it is more accurate and trustable.

Unsupervised Learning does not use any labelled data, it does the learning with the help of features present in the data.